



- **Problem Statement Title – Maximizing Section Throughput using AI-Powered Precise Train Traffic Control (Railways)**
- **Idea Title – Preventive Rail Traffic Control (PRTC)**
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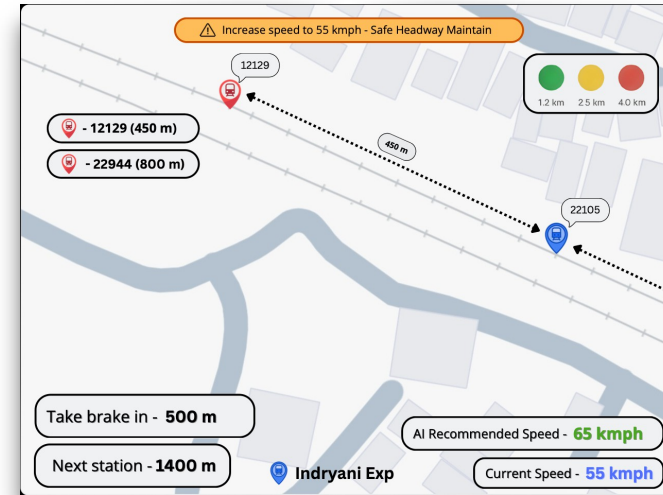
# IDEA TITLE: Preventive Rail Traffic Control (PRTC)

*AI-powered Preventive System for Safer, Smarter and Faster Train Operations.*

## Proposed Solution

- Presently two systems are used:
  - Absolute Block System** → Allows only one train between two stations.
  - Automatic Block System** → Multiple trains permitted, but must follow rigid, pre-defined block signals.
- Problem: Large sections of track remain unused even when they are safe for operation, leading to delays and inefficiency.
- Our Innovation – AI-Powered Control:**
  - Real-time tracking of train speed, position, length, and braking distance.
  - AI dynamically calculates safe headway and speed for each following train.
  - Enables multiple trains to run safely within one section using adaptive spacing.
  - Provides live speed and caution updates directly to the loco pilot and control room.
  - Transforms train control from reactive decision-making to predictive, preventive control.**

★ *Transforms rail control from reactive decision-making to predictive, preventive operation.*

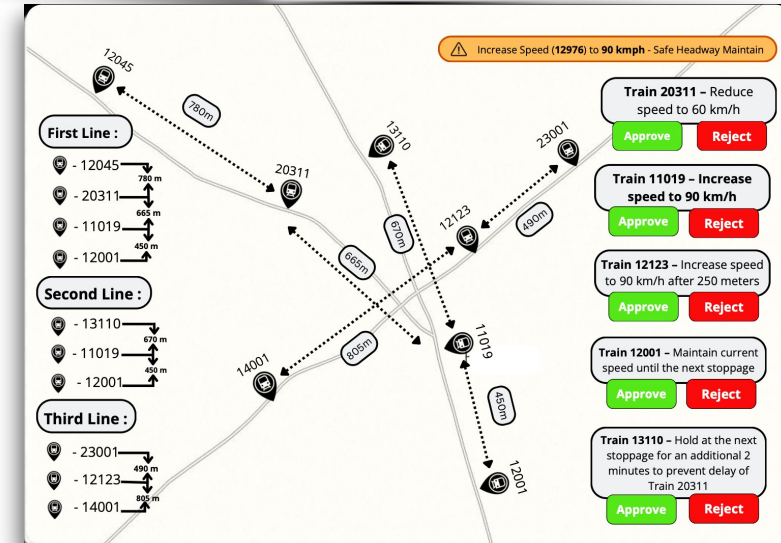


## RAILMATE -

*AI-powered smart assistant for locomotive pilots*

## RAILCOMMAND -

*AI-powered traffic control display for controllers*



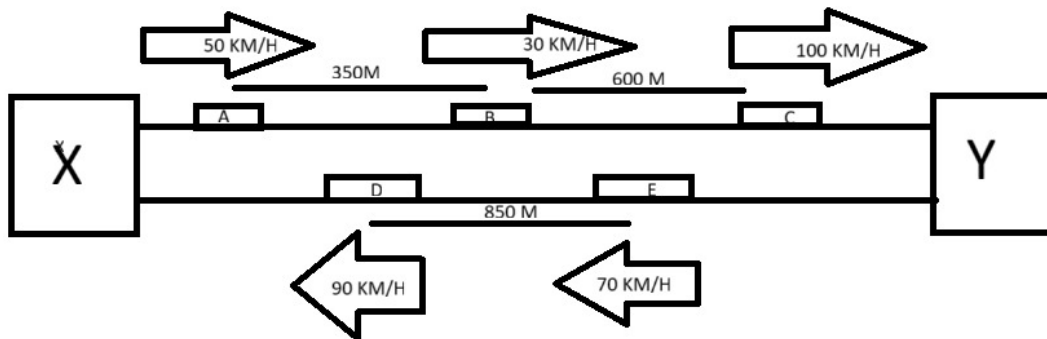
# TECHNICAL APPROACH

## Real-Time Data Collection

- Each train is equipped with a **smart onboard tablet** for data collection.
- Captures speed, braking type, train length, acceleration, and driver reaction time.
- Data is continuously synchronized with the central control system and loco pilot interface.

## Stopping Distance Calculation (Formulas Used):

1. **Reaction Distance**  $\rightarrow S_{reaction} = u \times t_{reaction}$ 
  - $u = \frac{speed (km/h) \times 1000}{3600}$
2. **Braking Distance**  $\rightarrow s = \frac{u^2}{2a}$
3. **Total Stopping Distance**  $\rightarrow S_{total} = S_{reaction} + s$



AI engine continuously computes safe braking distance using real-time formulas, adapting to train-specific conditions.

## AI-Powered Engine

- Calculates real-time **safe distance** using above formulas.
- Predicts potential conflicts and automatically suggests preventive actions (speed regulation or controlled halt).
- Adapts to different braking system (Air Brakes, Electro Pneumatic, Vacuum Brakes)
- Continuously **self-learns** and refines accuracy based on train-specific data.

## Technology Stack

- **Deep Reinforcement Learning (DRL):** Optimizes train speed profiles for maximum throughput.
- **Operations Research (OR) Models:** Ensures constraint handling for safety and headway.
- **Digital Twins:** Tests every decision virtually before real execution..
- **APIs:** Integrates GPS, signalling data, braking and IOT sensors in real-time.

## Visualization & Control

- **Loco Pilot Display:** Shows AI-recommended speed and safe following distance.
- **Control Dashboard:** Monitors all trains live with human override capability.
- **Human-in-loop:** Loco pilot follows AI guidance, but station master/control can adjust.

Data Collection  $\rightarrow$  AI Engine (DRL + OR Models)  $\rightarrow$  Simulation via Digital Twin  $\rightarrow$  Output: Loco Dashboard + Control Room

# FEASIBILITY AND VIABILITY

- **Feasibility**

- Utilizes existing GPS modules, tablets, and track circuits — no major infrastructure overhaul needed.
- Fully compatible with both Absolute and Automatic Block Systems.
- The AI engine functions as a **lightweight software layer** over existing signalling networks.
- **Cloud–Edge hybrid setup** for low-latency AI decisions even in remote zones.
- **Integrates with RailNet / COA / NTES data** for real-time coordination.
- **Fail-safe redundancy** ensures safe fallback to current signalling in case of system uncertainty
- **Simulation-validated through Digital Twin** before live trials.

- **Challenges & Mitigation**

- **Data Availability:** Begin with simulations → progressively integrate live Indian Railways data.
- **Safety Assurance:** Conservative AI decisions always validated by human operators.
- **Scalability:** Modular, corridor-wise design for phased expansion.

- **Viability**

- After successful pilots, system is **scalable to nationwide deployment**.
- Long-term adoption aligns with Government’s “**Digital Railways**” vision.
- Provides **low-cost digital enhancement** compared to expensive signalling upgrades.

- **Pilot Deployment**

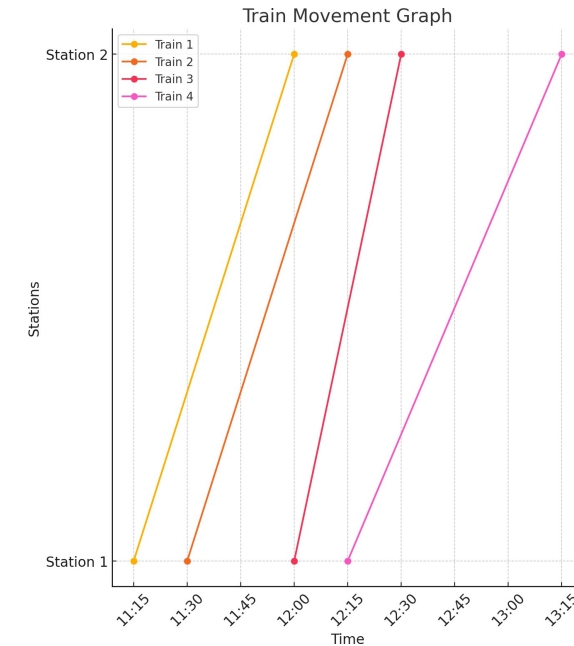
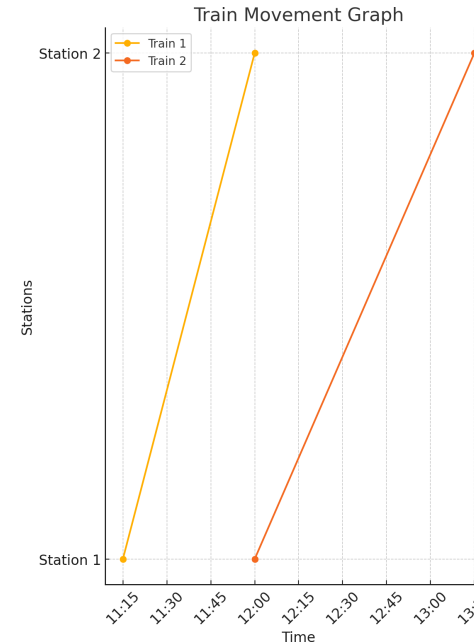
- Pilot rollout on **busy, high-density corridors** such as Delhi–Kanpur or Mumbai–Pune.
- Demonstrates safe, parallel operation of 3–5 trains within a section.

- **Key Features**

- Train-specific recommendations based on braking system (Air, Electro Pneumatic, Vacuum).
- Maintains **Human-in-loop architecture** for confirmed, **fail-safe operation**.

# IMPACT AND BENEFITS

- **Operational Impact**
  - **Increases section throughput** – enables more trains per track with complete safety.
  - **Reduces headway** intelligently, maintaining safety at all times.
  - **Improves punctuality and reliability** across passenger and freight trains.
- **Economic Benefits**
  - **Cuts delay-related losses** – saving fuel, crew hours, and energy.
  - Boosts Indian Railways' operational efficiency with minimal infrastructure cost.
- **Passenger & Social Benefits**
  - **Enhances passenger satisfaction** through better on-time performance and fewer cancellations.
  - Builds **public confidence** in Indian Railways as a modern, AI-driven transport network.
- **Environmental Benefits**
  - **Reduced fuel wastage** and **Lower Carbon emissions** by minimizing idle time and unnecessary acceleration.
  - Aligns with India's Green Mobility & Sustainability Mission.



- **Strategic Benefits**
  - A major step toward AI-driven Digital Railways transformation.
  - Technology built for nationwide scalability and interoperability.
  - **Future-ready foundation** for smart mobility and multimodal transport integration.

# RESEARCH AND REFERENCES

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Link : <https://rdso.indianrailways.gov.in/works/uploads/File/p%29%20irpwm-2004.pdf> 

**Topic : Braking Systems in Indian Railways – IJERT Journal**

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**Topic : Signalling Systems in Indian Railways – IRISSET Course Material**

Link : <https://iriset.railnet.gov.in/content/coursmat/cli/coumate/CLI-01.pdf>

**Topic : Indian Railways General & Subsidiary Rules, Chapter 8 – Absolute Block System**

Link : [https://indianrailways.gov.in/railwayboard/uploads/directorate/safety/CR/GR\\_SR\\_Chap-8\\_CR.pdf](https://indianrailways.gov.in/railwayboard/uploads/directorate/safety/CR/GR_SR_Chap-8_CR.pdf)

**Topic : Northern Eastern Railway Report – Automatic Intermediate Block Signalling**

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**Topic : RFCA FAQ – Train Working and Signalling Basics**

Link:- [https://www.irfca.org/faq/faq-signal4\\_b.html](https://www.irfca.org/faq/faq-signal4_b.html)